

permeability $\sim$ conductivity

$$V = IR$$

$$J = \sigma E$$

$$\frac{A}{m^2} \rightarrow \frac{V}{m}$$

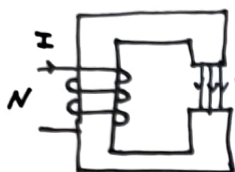
$$B = \mu H$$

$\frac{Wb}{m^2}, T$ $\frac{A}{m}$

$$R = \frac{l}{\sigma A}$$

$$R = \frac{l}{\mu A}$$

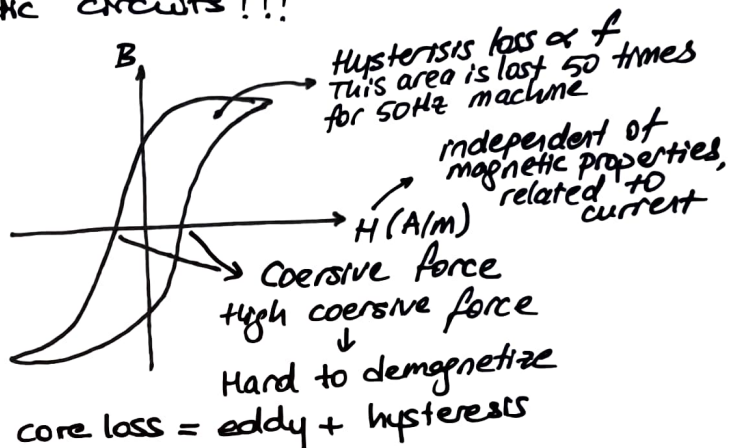
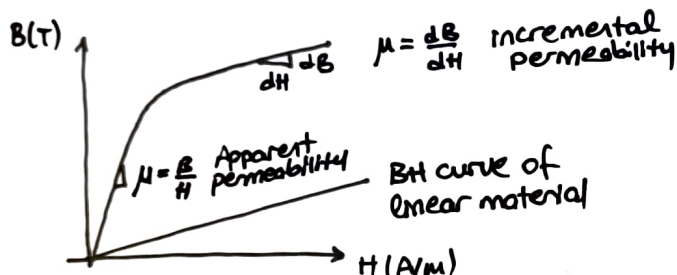
$$L = \frac{\lambda}{I} = \frac{N^2}{R}$$



$$NI = \Phi R$$

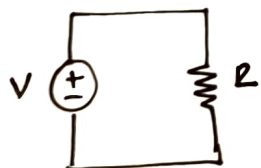
$\underbrace{NI}_{\text{Total MMF}} \rightarrow \text{reluctance}$
 $\rightarrow \text{flux}$

→ Air is a circuit element in magnetic circuits!!!



$$e = -N \frac{d\Phi}{dt} = -\frac{d\lambda}{dt} = -\frac{d(N\Phi)}{dt}$$

Eddy Current Loss



$$V = IR$$

$$P = \frac{V^2}{R}$$

$$\sigma E = J \propto I$$

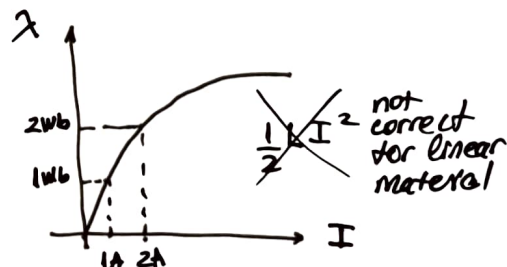
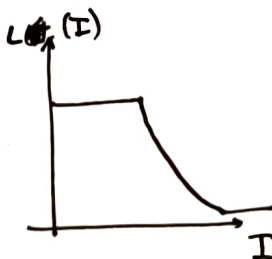
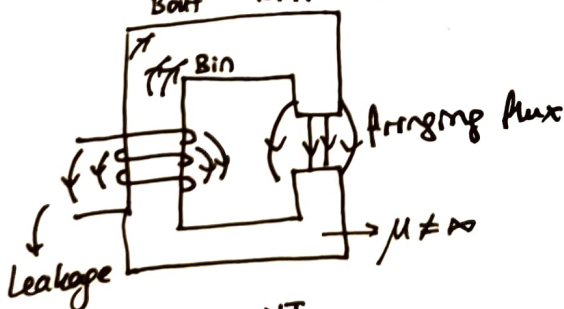
$$e = N \frac{d\Phi}{dt} \rightarrow B$$

$\underbrace{\frac{d\Phi}{dt}}_{f}$

$$P_{\text{eddy}} \propto B^2 \propto f^2 \propto \sigma$$

$\propto V^2$

$B_{in} > B_{out}$ due to lower reluctance $\Rightarrow$ Non-uniform flux distribution.



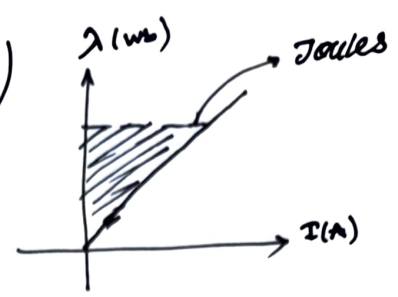
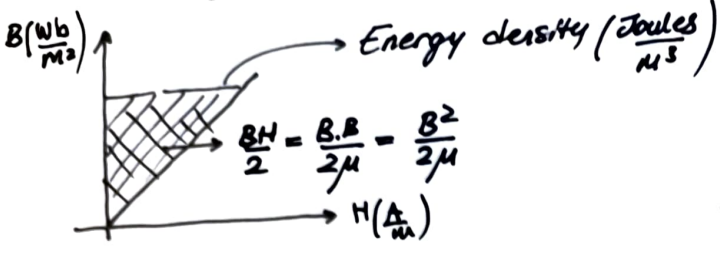
$$L = N \frac{d\Phi}{dI} = \frac{NI}{\frac{dI}{d\Phi}} = \frac{d\lambda}{dI}$$

second N term is inside the flux

$$L = \frac{N^2}{R}$$

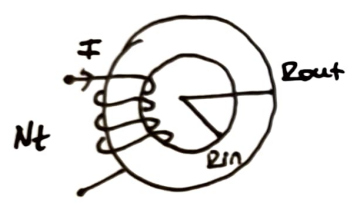
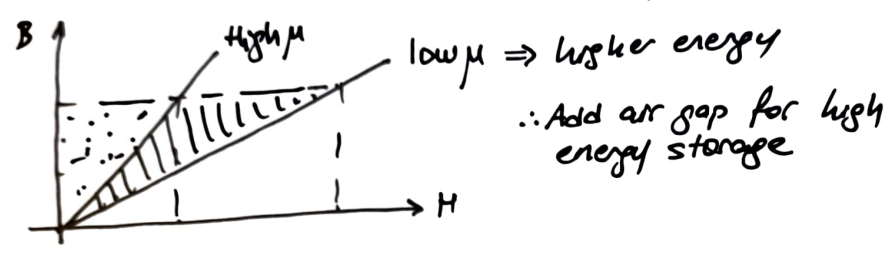
$$L = \frac{\lambda}{I} \rightarrow \text{for linear material}$$

Total Energy stored (Linear Systems)



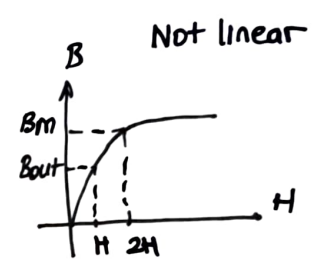
$W = \int H dB$: Energy density

$W = \frac{1}{2} \frac{\text{Volume}}{\mu} B^2$ stress tensor = $\frac{B^2}{2\mu_0}$ $\frac{\text{Joules}}{\text{m}^3} \rightarrow \frac{\text{N}}{\text{m}^2} \rightarrow \text{Pa}$

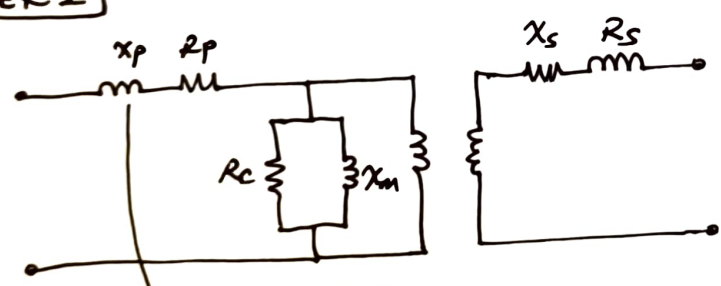


$\oint H dl = NI$
 $H \cdot 2\pi R_m = NI$
 $\frac{B}{\mu} 2\pi R_m = NI$

$R_{out} = 4\text{cm} \rightarrow H$
 $R_{in} = 2\text{cm} \rightarrow 2H$



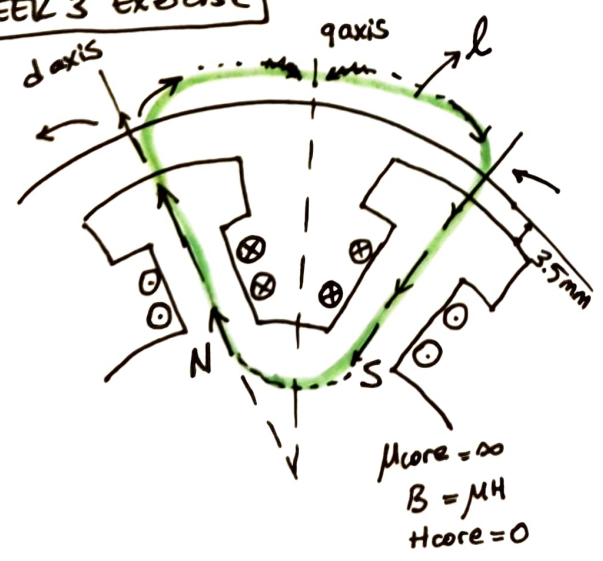
WEEK 2



Leakage inductance
 - air cored
 - more linear

$JWL = J\lambda_m + J\lambda_p$

WEEK 3 Exercise

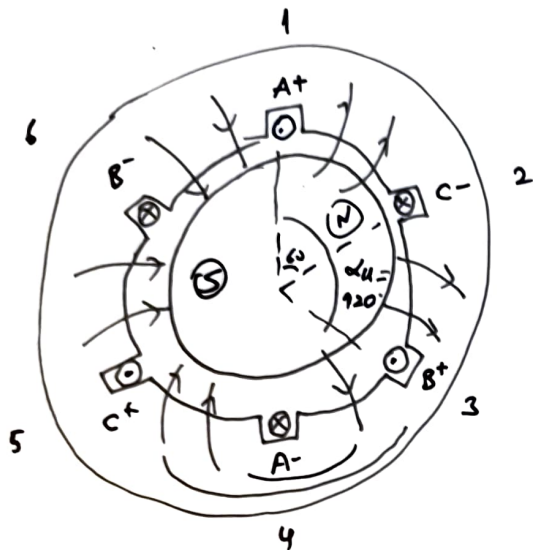


$\oint H dl = NI$
 $2(H_{core} l_{core} + H_{gap} l_{gap} + H_{back_{core}} + l_{back_{core}}) = NI$

$2H_{gap} l_{gap} = 2NI$

$\left(\frac{0.82T}{4\pi \times 10^{-7}} \right) (3.5\text{mm} \times 10^{-3}) = 95I$

$I \approx 2\text{mA}$



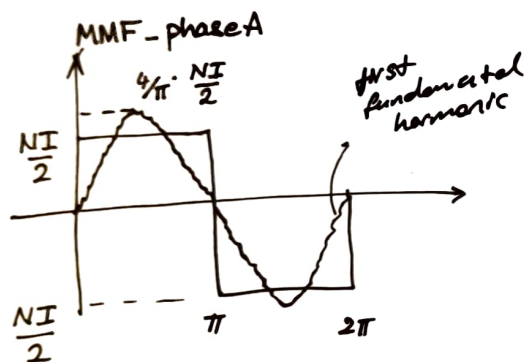
$m = 3$: # of phases

$p = 1$: pole pair

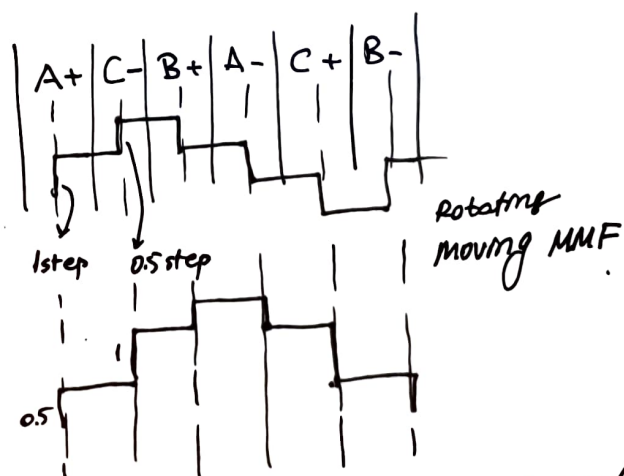
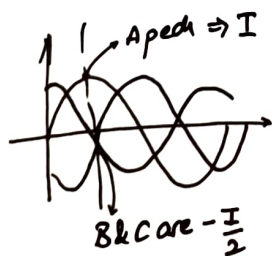
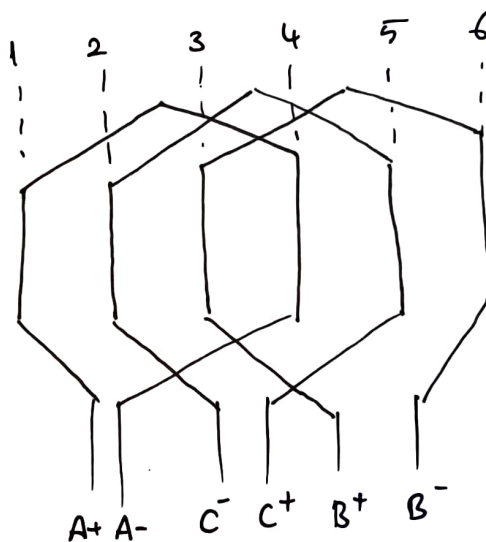
$q = 1$: slots per pole per phase

$$\alpha_u : \frac{2\pi}{Q/p} = \frac{360^\circ}{6/1} (\text{slot angle}) = 60^\circ$$

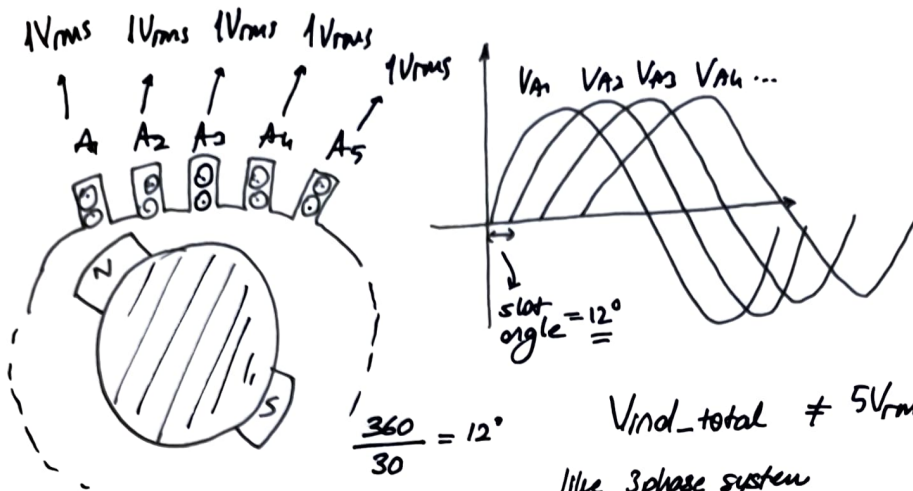
# of slots	1	2	3	4	5	6
phase	A+	C-	B+	A-	C+	B-



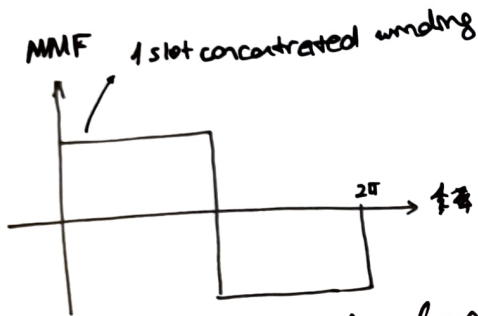
$$\text{peak mmf} = \frac{4}{\pi} \times \frac{NI}{2}$$



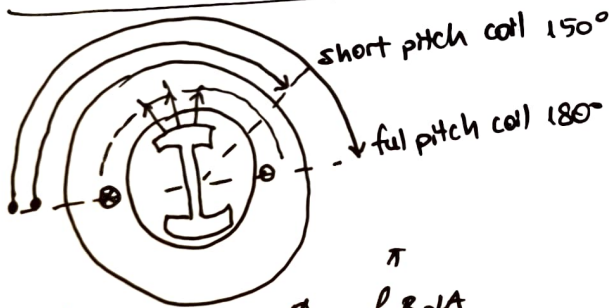
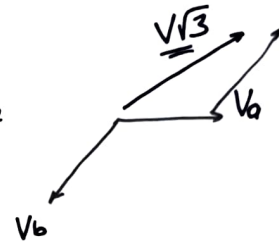
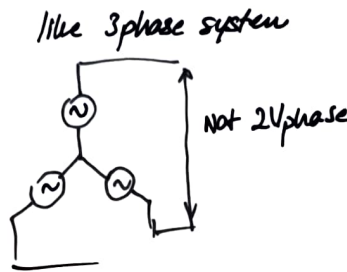
low slot number $\rightarrow$ discrete mmf
more harmonics?
or different harmonics?
 $\hookrightarrow$ obvious



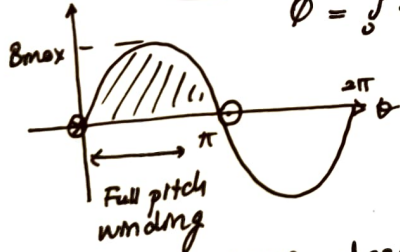
$V_{ind_total} \neq 5V_{rms}$ → we have the phase difference!



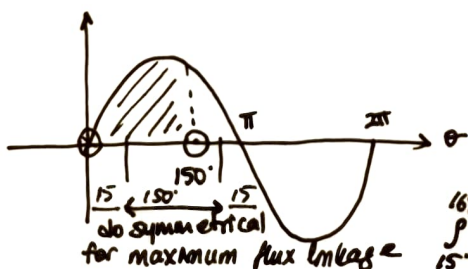
stepped waveform for distributed windings



$$\Phi = \int_0^\pi B dA$$



180° electrical degree between $A+$ & $A-$ windings.



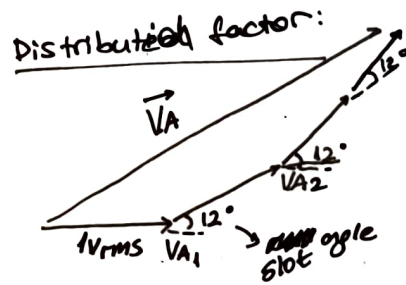
short pitch winding $< 180^\circ$

"5/6 pitched"

$$\int_0^\pi B dA = \Phi$$

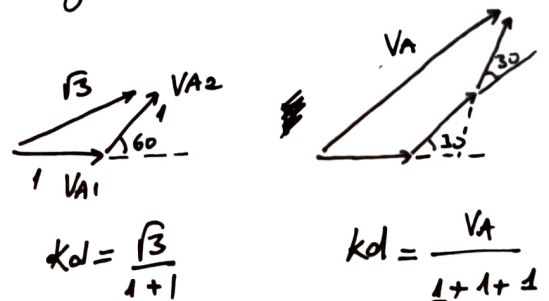
► Distribution factor (k_d): Due to distributed winding

► Pitch factor (k_p): Due to short / full pitch



$$V_A < 5V$$

$$k_d = \frac{\text{Vector sum of voltages}}{\text{Algebraic sum of voltages}} < 1$$



$$k_d = \frac{\sqrt{3}}{1+1}$$

$$k_d = \frac{V_A}{1+1+1}$$

Generalised k_d :

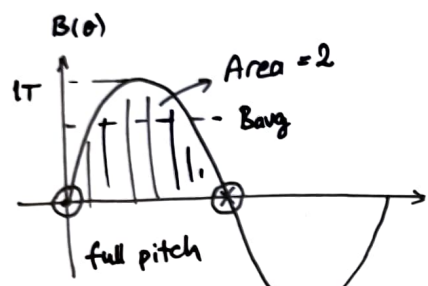
$$k_d = \frac{\sin(q \frac{\alpha}{2})}{q \sin(\frac{\alpha}{2})}$$

q : number of coils

α : angle between each coils
slot angle (electrical)

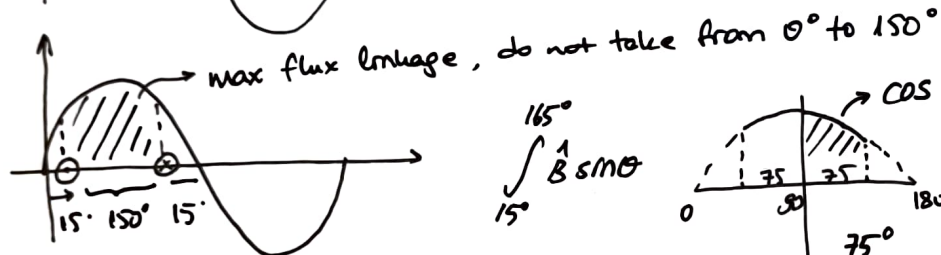
short pitched $\rightarrow$ some harmonics are eliminated

k_p : $\frac{V_{\text{short pitched coil}}}{V_{\text{full pitched coil}}}$
 $\uparrow$
pitch factor

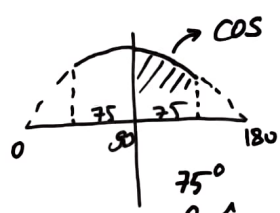


$$\Phi_{\text{max}} = l \int_0^\pi B(\theta) d\theta \quad \leftarrow \text{max flux linkage}$$

$$B_{\text{avg}} = \frac{2}{\pi} \cdot \hat{B}_{\text{peak}}$$



$$\int_{15^\circ}^{165^\circ} \hat{B} \sin \theta d\theta$$



$$\int_0^\pi \hat{B} \cos(\theta) d\theta$$

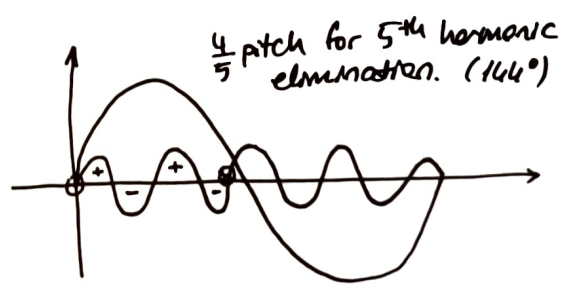
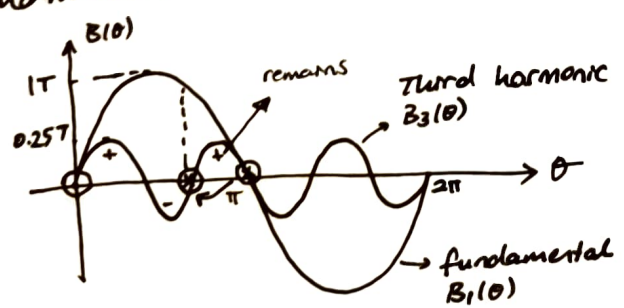
$$\text{Area} = 2 \times \sin(75^\circ) \quad \text{for } \hat{B}=1$$

Pitch factor: $k_p = \sin(\frac{\lambda}{2})$

Why we use short pitch then?

$\rightarrow$ Filtering harmonic components

Even harmonics means unbalance, DC bias. We have N & S poles, in magnetic field distributions we have both + & - cycles which eliminates 2nd or even harmonics. We cannot obtain DC bias.



- We have 3rd harmonic for full pitch winding $\rightarrow$ this is not the best way
- For $\frac{2}{3}$ pitched winding (120°) $\rightarrow$ No third harmonic voltage is induced!

Why we want to eliminate harmonics? $\rightarrow$ In general no effect on torque creates losses.

~~$k_p(n^{th} \text{ harmonic}) = \sin(\frac{n\lambda}{2})$~~ ~~$k_d = \sin(\frac{n\alpha}{2})$~~

$k_p(n^{th} \text{ harmonic}) = \sin(\frac{n\lambda}{2})$ n : harmonic order $\therefore$ pitch factor

$k_d(n^{th} \text{ harmonic}) = \frac{\sin(nq\frac{\alpha}{2})}{q\sin(\frac{n\alpha}{2})}$ n : harmonic order $\therefore$ distribution factor

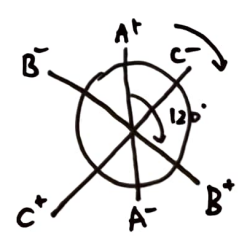
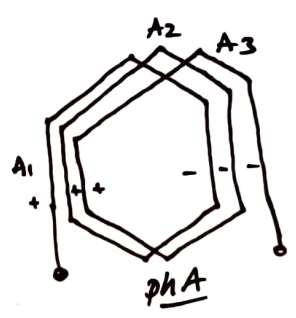
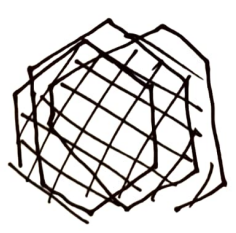
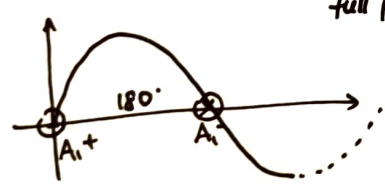
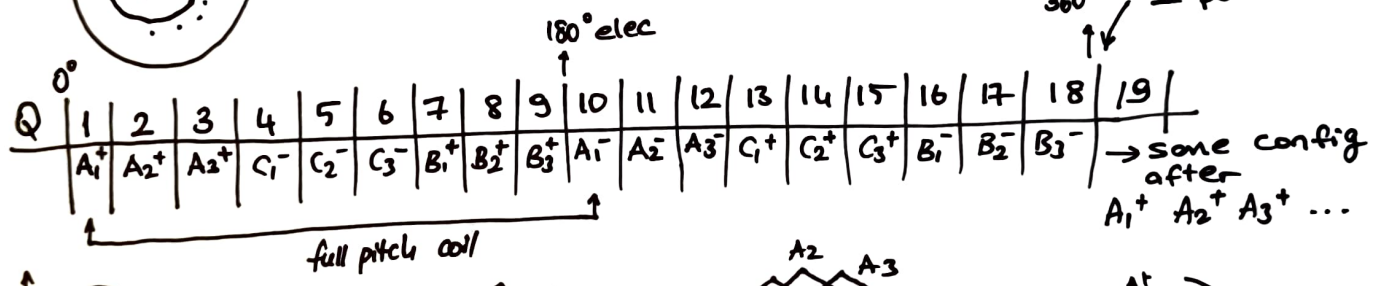
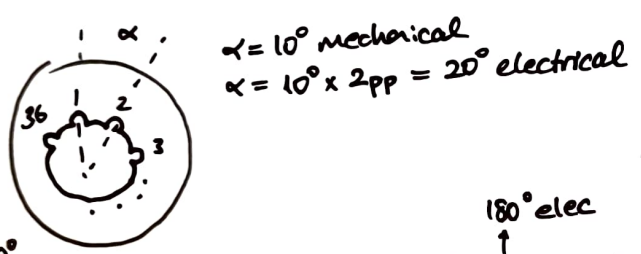
winding factor:

$k_w = k_p \times k_d$

Hard to physically wound the windings for short pitched coil.

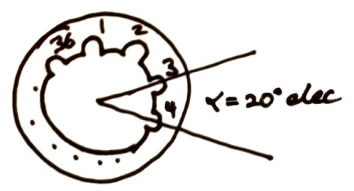
36 slots, 3 phase, 4 poles

$q = \frac{36}{4 \times 3} = 3$ slots per pole per phase



↑ Full pitched coil winding diagram.

short pitched winding



Q	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
	A ₁ ⁺	A ₂ ⁺	A ₃ ⁺	C ₁ ⁻	C ₂ ⁻	C ₃ ⁻	B ₁ ⁺	B ₂ ⁺	A ₁ ⁻	A ₂ ⁻	A ₃ ⁻	C ₁ ⁺	C ₂ ⁺	C ₃ ⁺	B ₁ ⁻	B ₂ ⁻	B ₃ ⁻	
									B ₃ ⁺									

we should
no double
layer.

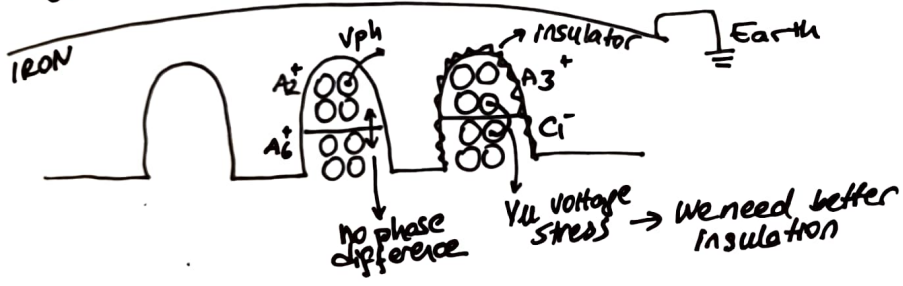
$\alpha = 120^\circ \rightarrow$ we cannot choose 150° short pitch
 160° short pitched
 $\frac{8}{9}$ pitched $\rightarrow$ 8 spiri:



Q	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
	A ₁ ⁺	A ₂ ⁺	A ₃ ⁺	C ₄ ⁻	C ₅ ⁻	C ₆ ⁻	B ₁ ⁺	B ₂ ⁺	B ₃ ⁺	A ₄ ⁻	A ₅ ⁻	A ₆ ⁻	C ₁ ⁺	C ₂ ⁺	C ₃ ⁺	B ₄ ⁻	B ₅ ⁻	B ₆ ⁻
	A ₄ ⁺	A ₅ ⁺	A ₆ ⁺	C ₁ ⁻	C ₂ ⁻	C ₃ ⁻	B ₄ ⁺	B ₅ ⁺	B ₆ ⁺	A ₁ ⁻	A ₂ ⁻	A ₃ ⁻	C ₄ ⁺	C ₅ ⁺	C ₆ ⁺	B ₁ ⁻	B ₂ ⁻	B ₃ ⁻

tetris
schür

go with bottom layer for equal length



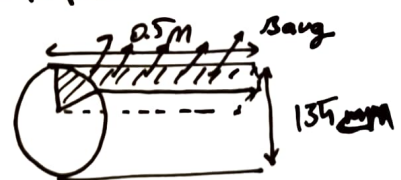
- ⊛ production problems
- ⊛ removing burned out
phases are problematic
(eliminating from the slots
is very hard)
- ⊛ But harmonic elimination
is a plus!

Exercise: (slide 54)

- $\rightarrow$ 180 slots, double layer, Y connected
- $\rightarrow$ Each coil has 3 turns with a span of 15 slots
- $\rightarrow$ Armature dia = 135cm, eff axial length = 0.5m
- $\rightarrow$ A rotating machine flux created by 3ph 50Hz, 600ram alternator.

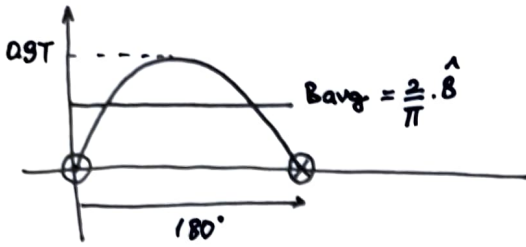
$$N_s = \frac{120f}{P} = \frac{120 \cdot 50}{P} = 600 \Rightarrow P = 10 \text{ poles}$$

$$q = \frac{180 \text{ slots}}{3 \text{ phase} \times 10 \text{ poles}} = 6$$



$$\text{pole area} = \frac{\pi D a L}{2 p p} = \frac{\pi \times 1.35 \times 0.5}{10} = 0.212 \text{ m}^2$$

$$\rightarrow B(\theta) = 0.9 \sin \theta + 0.25 \sin(3\theta) + 0.15 \sin(5\theta)$$



$$\phi = ?$$

$$\phi = \left(\int_0^\pi B \cdot d\theta \right) \cdot l$$

$$\hat{\phi}_1 = \frac{2}{\pi} \hat{B}_1 \cdot A_{pole} = \frac{2}{\pi} \times 0.9 \times 0.212$$

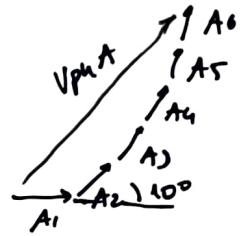
$$\hat{\phi}_1 = 12.4 \text{ mWb}$$

mWb is good ✓

We have calculated flux linkage for 1 turn concentrated winding, full pitch.

$$q=6 \quad \alpha = \left(\frac{360}{180} = 2^\circ \text{ mech} \right) \times 5 = \underline{10^\circ \text{ electrically}} = \pi/18$$

$$\frac{360^\circ \times 5}{180}$$



$$k_{d1} = \frac{\sin\left(\frac{9\alpha}{2}\right)}{9\sin\left(\frac{\alpha}{2}\right)} = \frac{\sin\left(\frac{6 \times 10^\circ}{2}\right)}{6\sin\left(\frac{10^\circ}{2}\right)} = \frac{\sin(30)}{6\sin(5)} = \underline{0.956} \quad \text{we only lost 5\%}$$

$$k_{d3} = \frac{\sin(90)}{6\sin(15)} = \underline{0.643} \quad \text{do not eliminate 3rd harmonic, but reduced it.}$$

$$k_{d5} = \frac{\sin(150)}{6\sin(25)} = \underline{0.197} \quad \text{eliminated ~~80\%~~ 80\%}$$

$$(\alpha = 10^\circ \text{ degrees}) \times (15 \text{ slots}) = 150^\circ$$

$$\lambda = 150^\circ \text{ coil pitch}$$

$$\lambda = \frac{5\pi}{6} \text{ ~~per pole~~}$$

Try to do the exercises at home! until wednesday!

Group exercise (slide 54)

- A rotating magnetic flux created by 3 phase 50Hz, 600rpm alternator has a distribution of magnetic flux density:

$$B(\theta) = \hat{B}_1 \sin(\theta) + \hat{B}_3 \sin(3\theta) + \hat{B}_5 \sin(5\theta)$$

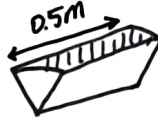
$$B(\theta) = 0.9 \sin(\theta) + 0.25 \sin(3\theta) + 0.18 \sin(5\theta)$$

- 180 slots, double layer winding, Y connected
- Each coil has 3 turns with a span of 15 slots
- Armature diameter 135cm
- Effective axial length = 0.5m

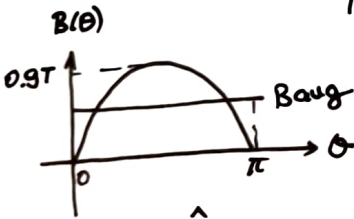
$$(f_m = \frac{600 \text{ rpm}}{60} = 10 \text{ Hz}) \times p_p = f_{\text{elec}} = 50 \text{ Hz} \Rightarrow p_p = 5 \Rightarrow \underline{10 \text{ pole}}$$

$$q = \frac{180}{10 \cdot 3} = 6 \text{ slots per pole per phase}$$

$$A_{\text{pole}} = \frac{\pi D_{\text{armature}} \times L}{(2 \times p_p)}$$



$$= \frac{\pi (1.35 \text{ m}) (0.5 \text{ m})}{10} = 0.212 \text{ m}^2$$



$$\Phi_1 = \int_0^\pi \hat{B} dA = \int_0^\pi (0.9 \sin \theta) d\theta \cdot L$$

$$\text{veya } \hat{\Phi}_1 = B_{\text{avg}} \cdot A_{\text{pole}} = \frac{2}{\pi} \hat{B} \times A_{\text{pole}} = \frac{2}{\pi} \hat{B} \left(\frac{\pi D L}{2 p_p} \right)$$

$$B_{\text{avg}} \cdot \pi = \hat{B} \times 2$$

$$B_{\text{avg}} = \frac{2}{\pi} \hat{B}$$

$$= \frac{2}{\pi} \hat{B} \frac{\pi r l}{p_p} = 2 \hat{B} \frac{r l}{p_p}$$

$$\hat{\Phi}_1 = \int_0^\pi \hat{B} dA = \int_0^\pi \hat{B} r dl d\theta_m$$

$$\text{either } \pi \text{ to mechanical or } \pi \text{ to electrical} = \int_0^\pi \hat{B} r dl d\theta_e \cdot \frac{2}{p_{\text{pole}}} = (0.9 \times 2) \left(\frac{1.35}{2} \right) (0.5) \cdot \frac{2}{p_{\text{pole}}}$$

$$= 2 \cdot \hat{B} \cdot \frac{r \cdot l}{p_p} \Rightarrow \boxed{\hat{\Phi}_1 = 121.4 \text{ mW}}$$

$$\alpha : \text{slot angle} = \left(\frac{360}{180} = 2 \right) \times \frac{10 \text{ pole}}{2} = \underline{10^\circ} \text{ electrical}$$

$$\lambda : \text{coil pitch} = \underline{\quad}$$

$$180 \text{ slot} \rightarrow 10 \times 180^\circ \text{ electrical}$$

$$15 \text{ slot} \rightarrow \frac{10 \times 180 \times 15}{180} = 150^\circ \text{ electrical} = \lambda$$

$$\lambda = 15 \times \alpha$$

$$\text{True RMS} = \sqrt{V_1^2 + V_3^2 + V_5^2}$$

$$V(t) = 4482\sqrt{2} \sin(x) - 613\sqrt{2} \sin(3x) + 49.5\sqrt{2} \sin(5x) \leftarrow \text{sharp sinusoid}$$

Y connected $\rightarrow$ 3rd harmonic elimination

for VLL voltage $\Rightarrow$ 1st + 5th harmonic $\Rightarrow$ more sinusoidal shape

Group exercise continued:

	1st	3rd	5th	calculated before
k_d	0.956	0.643	0.197	
k_p	0.966	-0.707	0.259	
k_w	0.923	-0.4546	0.051	

$$k_p = \sin\left(\frac{n\lambda}{2}\right)$$

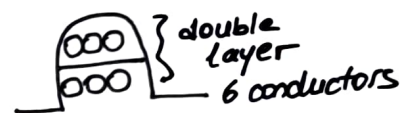
$-k_p \rightarrow 180^\circ$ phase shift
reverses rotation direction as it changes phase sequence.

$$k_w = k_p \cdot k_d$$

$$q = 6 \frac{\text{slots}}{\text{pole} \cdot \text{phase}}$$

$$N \left(\frac{\text{turns}}{\text{phase}} \right) = \underbrace{\frac{180 \text{ slots}}{3}}_{\text{slots per phase}} \times 6 \div 2 = 180 \frac{\text{turns}}{\text{phase}}$$

$$\text{or } (q=6) \times 10 \text{ pole} \times 3 \text{ turns} = 180 \text{ turns/phase}$$

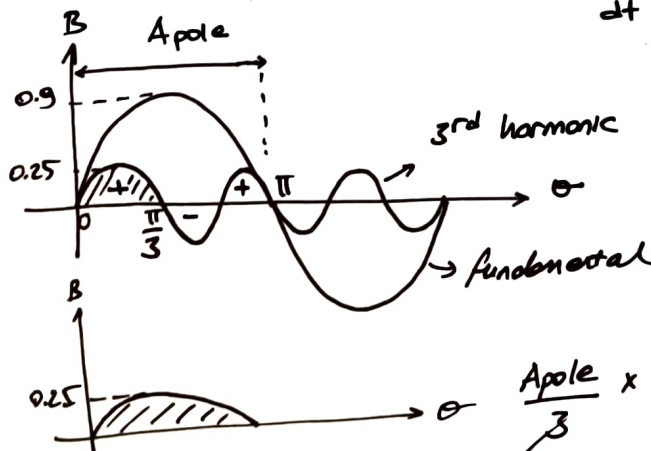


$$V_{\text{ind, phase}} = \frac{2\pi}{\underbrace{12}_{4.44}} f N \underbrace{B}_{\text{avg } \phi} A_{\text{pole}} \cdot k_w$$

$$V = N \frac{d\phi}{dt} = N \frac{dB}{dt} \text{ where } B = \hat{B} \sin \theta$$

$$\frac{dB}{dt} = \hat{B} \cos(2\pi ft) \cdot 2\pi f$$

$$\left. \begin{array}{l} V_1 = 4480V \\ V_3 = -613V \\ V_5 = 49.5V \end{array} \right\} \text{rms}$$



$$\frac{A_{\text{pole}}}{3} \times 3f \text{ for 3rd harmonic}$$

or A_{pole} is same then B_{avg} is reduced to $1/3$

$$B_{\text{avg}} \cdot \pi = \int_0^{\pi/3} \hat{B} \sin(3\theta) d\theta \Rightarrow \frac{\hat{B}}{3} \cdot 2 = B_{\text{avg}} \cdot \pi \Rightarrow B_{\text{avg}} = \frac{2\hat{B}}{3\pi}$$

Fractional slot windings

q is fractional, not integer!

Exp: 10 pole machine

30 slots $\rightarrow q=1$
 60 slots $\rightarrow q=2$ } every pole is symmetrical $\rightarrow$ we can connect poles series or parallel.

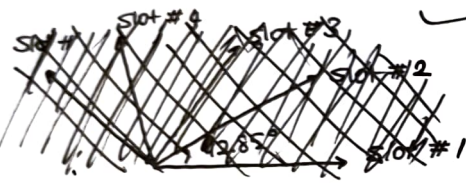
42 slots $\rightarrow q=7/5 \rightarrow$ fractional slot! $\rightarrow$ Vind between poles are not identical we cannot connect them in parallel!

$$\frac{42 \text{ slots}}{3 \text{ phase}} = 14 \text{ slots/phase}$$

$$\frac{44 \text{ slots}}{3 \text{ phase}} \Rightarrow \text{not integer, not balanced!}$$

* # of slots should be multiple of slots for balanced machine.

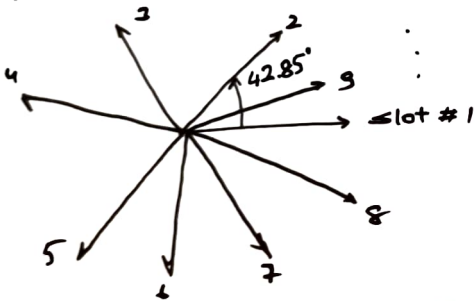
$$\alpha_n = ? \quad 42 \text{ slots} \Rightarrow \frac{360^\circ \text{ mechanical}}{42} \times 5_{pp} = 42.85^\circ \text{ electrical}$$



Lets chose a coil span of 4 slots coil pitch

$$42.85 \times 4 = 171.5^\circ$$

$$\text{Pitch factor} = \sin\left(\frac{171.5}{2}\right) \leftarrow \text{check.}$$



Why fractional winding? Cogging torque! $\rightarrow$ skew winding $\rightarrow$ increases R_{phase} to reduce it
 (torque ripple due to teeth!)

$\rightarrow$ 4 pole, 12 slots $\rightarrow q=1$
 For fractional slot, cogging torque is not added but symmetry is removed

14 pole 12 slots
 10 pole 12 slots $\rightarrow$ multiples are also used commonly

$\rightarrow$ E-BVL is a better ~~at~~ option for fractional slot

$\blacktriangleright$ 2 pole - 3phase - 12 slots $\rightarrow$ integer slot

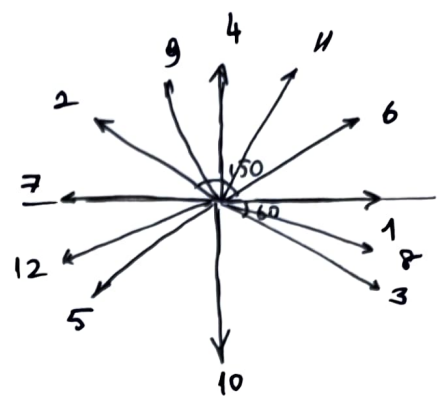
$$\alpha = \frac{360}{12} \times 1 = 30^\circ \text{ electrical}$$

$$q = \frac{12}{2 \times 3} = 2 \frac{\text{slots}}{\text{pole} \cdot \text{phase}}$$

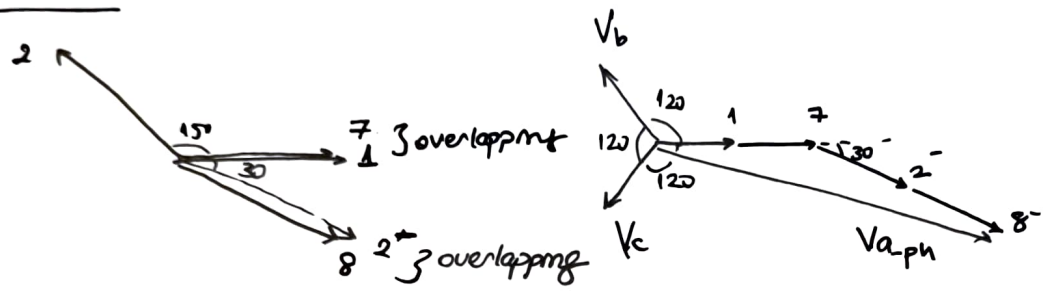
Example :

12 slots, 10 pole, single layer

$$\frac{360 \times 5pp}{12 \text{ slot}} = 150^\circ \text{ electrical}$$



Phase A



Example (2-16) $Q=36 \text{ slots}, pp=2, m=3$

$$\alpha = \frac{360 \times 2pp}{36 \text{ slots}} = 20^\circ \text{ electrical angle}$$

$$q = \frac{36}{4.3} = 3$$

work on slides !!!

0-18 slot } identical
19-36 slot }

(coil 1 & 19 are identical overlapping.)

$$\rightarrow 6 \text{ pole } 6 \text{ slot} \Rightarrow \frac{360 \times 3pp}{6 \text{ slot}} = 180^\circ$$

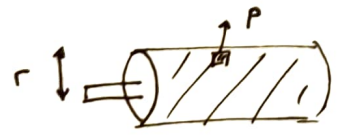
$$\frac{2}{4} \Rightarrow \frac{1}{2} \Rightarrow \frac{1}{5} \rightarrow 120^\circ \text{ phase shift is not possible}$$

check table 3.4 $\rightarrow$ combinations!!! check the book!!!

Design Procedures

maxwell stress tensor: $\sigma_F = \frac{\mu_0 H^2}{2} = \frac{B^2}{2\mu_0}$

Shear stress: $P_a = \frac{N}{M^2}$



$$\oint P dA \rightarrow F \times r \rightarrow T$$

Specific magnetic loading (T): average airgap flux density over a pole

$$\bar{B} = \frac{P\Phi_p}{\pi D L}$$

machine size $\uparrow$
cooling gets better
average shear stress $\uparrow$

$$\bar{A} = \frac{N_{\text{turn, slot}} I Q}{\pi D_i}$$

Specific Machine Constant (c)

$$S = m E_m I_s$$

↑
 KVA
 ↓
 # of
 phases
 ↓
 phase rms current
 ↓
 phase voltage

$$S_i = m \underbrace{\frac{1}{\sqrt{2}} \omega N_s k W_1 \Phi_m I_s}_{4.44 f N_s k W_1 \Phi_m \Rightarrow E_m}$$

$$S_i = m \frac{1}{\Gamma_2} \omega \cancel{N_s} \cancel{k_w} \hat{\Phi}_m \frac{\cancel{ATD}}{2mN_s}$$

flux per pole

$\text{A} = \frac{2\pi Ns}{\pi D} \cdot I_s$
 due to turn total num of conductors
 $\frac{\text{A}}{\text{M}}$

$$S_1 = \frac{1}{\sqrt{2}} \cdot 2\pi \phi n_{\text{syn}} \cdot k w_1 \cdot \frac{\pi D}{2\phi} \cdot \frac{2}{\pi} \hat{B} L' \frac{A \pi D}{2}$$

we cannot make a machine more powerful by increasing the pole!

$$S_i = \frac{\pi^2}{12} n_{\text{spin}} k_B A \hat{B} D^2 \lambda$$

► To increase power rating, increase;
Volume
Electric & Magnetic loading
Speed (synchronous)

$$S_i = C D^2 L' n_{syn}$$

$$C = \frac{\pi^2}{2} k W_1 A \hat{B}$$

C: specific machine constant
↳ performance evaluation regardless of the machine size ~~and~~

Example (Phy 6.1)

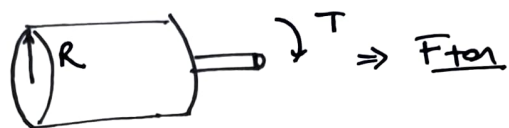
Dia = 98 mm Power = 4 kW pole = 2
 length = 82 mm f = 50 Hz slip (rated) = 7.1%

$P = T \omega_{\text{mech}}$
 $50 \text{ Hz} \times 60 = 3000 \text{ rpm}$ $s = 0.01$
 $2\pi f (1-s) = 99\pi$

$\frac{P = 4 \text{ kW}}{99\pi} = T = 12.86 \text{ Nm}$ SI unit !!!
 $\downarrow$
 10^{-6}

Rotor area $\Rightarrow S_r = \pi D r \cdot l = \pi 98 \text{ mm} 82 \text{ mm} \times 10^{-6} = 0.0252 \text{ m}^2$

$F_{\text{tor}} = \frac{12.86}{0.049} = 262 \text{ N}$



$F_{\text{tor}} = \frac{\text{torque}}{\text{radius}}$

Stress $\sigma = \frac{F_{\text{tor}}}{S_r} = \frac{262 \text{ N}}{0.0252} = 10.400 \text{ N/m}^2 \approx 10 \text{ kPa}$ (reasonable industrial motor)

$P_{\text{mech}} = C_{\text{mech}} D^2 L f_{\text{sync}}$

$C_{\text{mech}} = \frac{4000 \text{ W}}{0.098^2 \times 0.082 \times 50} = 102 \frac{\text{kJ s}}{\text{J m}^3}$ } Energy density like

Aspect Ratio

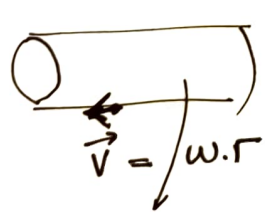
$\chi = \frac{L'}{D}$

Asynchronous machine

$\chi \approx \frac{\pi}{2p} \sqrt[3]{p}$

Synchronous machine

usually $0.5 < \frac{D_i}{L} < 2.5$



$r = 1 \text{ m}$ $N_r = 20.000 \text{ rpm} \Rightarrow \omega \approx 2000$

$v = \omega \cdot r \approx 2000 \text{ m/s}$

sound speed $\approx 300 \text{ m/s}$ Not feasible

$F_{\text{centrifugal}} = \frac{\omega^2 r \cdot m}{a}$

$a = \omega^2 r = 4 \cdot 10^6 \text{ m/s}^2 = 40.000 \text{ G}$

High speed cannot be possible with high radius

Example:

30kW, 690V, 50Hz, 4 pole, 3 phase, Squirrel cage Induction Generator (SCIG)
 choose: $C_{mech} = 150 \text{ kW/m}^3$

$$X' \approx \frac{L'}{D} = \frac{\pi \sqrt[3]{\rho}}{2p_{\text{pole pairs}}} = \frac{\pi \sqrt[3]{2}}{2 \cdot 2} = 0.989 \approx \frac{1}{7}$$

$$P_{mech} = C_{mech} \cdot \underbrace{D^2 L}_{\text{Volume}} f_{sync} \quad \text{as } D \approx L \Rightarrow D^3$$

$$D = \sqrt[3]{\frac{P_{mech}}{C_{mech} f_{sync}}} = \sqrt[3]{\frac{30 \text{ kW}}{150 \times 25}} \Rightarrow \boxed{D \approx 200.7 \text{ mm}} \approx 200 \text{ mm}$$

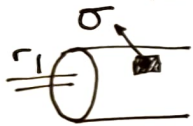
↑
mechanical

$$L = 1986 \text{ mm} \approx 200 \text{ mm}$$

Find the tangential stress if speed is 1474 rpm.

$$T = \frac{P}{\omega} \Rightarrow \frac{30 \text{ kW}}{2\pi \times \frac{1474}{60}} = 194.4 \text{ Nm}$$

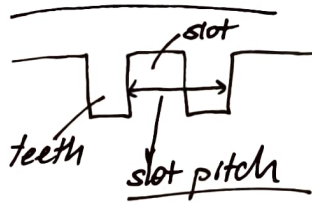
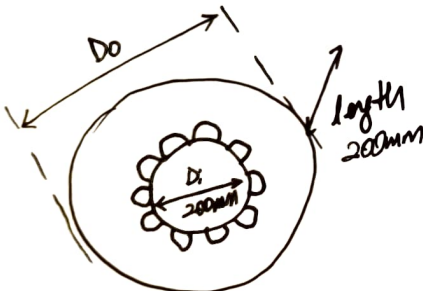
↑
rad/s



$$T = \sigma_{tan} \cdot A_{rotor} \cdot r \Rightarrow \sigma_{tan} = \frac{T}{A_{rotor} \cdot r} = \frac{194.4 \text{ Nm}}{\pi D L \cdot \frac{D}{2}}$$

(Rotor)
↓
surface

$$\sigma_{tan} = \frac{194.4}{\pi \left(\frac{0.2}{2}\right)^3} \approx 15.5 \text{ kPa}$$



Stator circumference: 628mm

Assume $q=2$, $N_{slot} = 2 \times 4 \times 3 = 24 \text{ slots}$

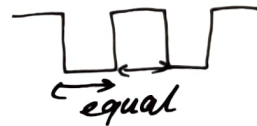
$$\text{tooth thickness: } T_{teeth} = \frac{628}{24 \times 2} = 13 \text{ mm}$$

for $q=4$, $N_{slot}=48$, $T_{teeth} = 6.5 \text{ mm}$

for $q=3$, $N_{slot}=36$, $T_{teeth} = 8 \text{ mm}$ ✓ - good choice.

coil span = ~~24~~ 9

Assume $W_{slot} = W_{teeth}$



MECHANICAL CONSTRAINTS

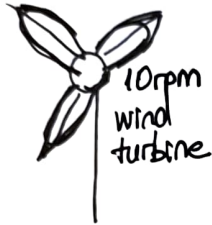
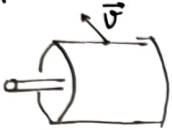
Previous example continued: what should be the air gap for 30kw heavy duty motor?

$$\delta = 1.6 \times 0.18 \times 0.006 (30\text{kw})^{0.4} = 0.88\text{mm} \approx \underline{0.9\text{mm}}$$

air gap $\rightarrow$ increase %60 because heavy duty

Tip speed:

What is the rotational speed of a machine with 0.5m rotor dia to reach tip speed of the sound (1 mach)?



$$V = \omega r$$

$$345 = \omega \cdot (0.25)$$

$$\omega = 1400 \text{ rad/s}$$

$$N \approx \underline{14,000 \text{ rpm}}$$



Example 6.3:

Calculate max diameter ϕ for a smooth steel cylinder having a small bore.
Speed 15000 rpm, Yield strength 300MPa. $\rho = 7860 \text{ kg/m}^3$.

Poisson ratio, around 0.3 for all metals

$$C' = \frac{(3+\nu)}{4}$$

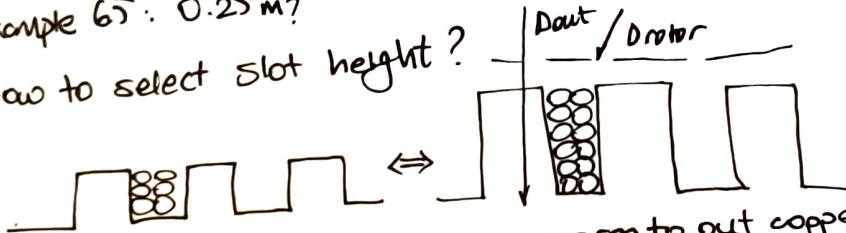
$$C' = \frac{(3+0.29)}{4} = 0.823$$

$$\sigma_{\text{mech}} = C' \rho (r^2) \omega^2 < 300 \text{ MPa}$$

$$r_{\text{max}} = \sqrt{\frac{300 \text{ MPa}}{(0.823)(7860) \left(\frac{2\pi \cdot 15000}{60}\right)^2}} \approx \underline{0.13 \text{ m}}$$

Example 6.5: 0.25m?

How to select slot height?



- more area to put copper $\rightarrow$ high electric loading
- cooling becomes an issue
- If Drotor is constant $\rightarrow$ larger machine
- If Dout is constant $\rightarrow$ smaller torque

* Example: $\rightarrow$ slot height

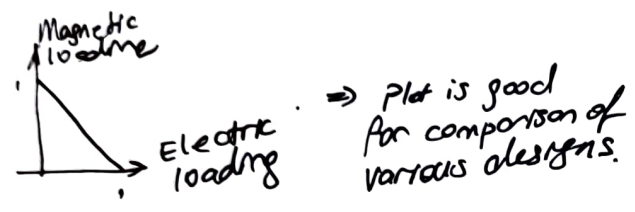
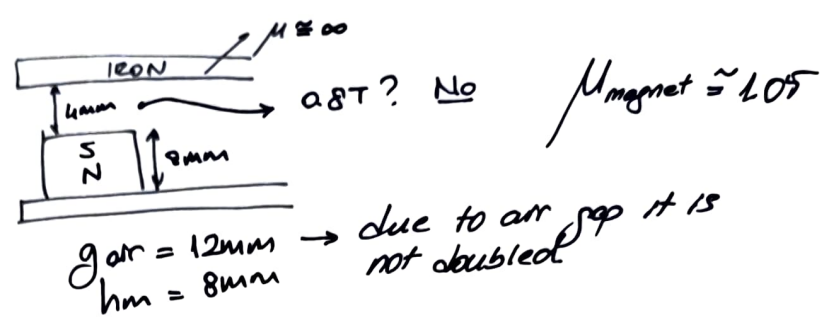
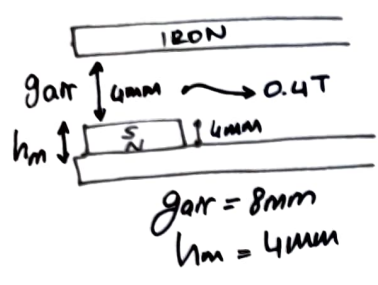
$$I \propto (1-d)$$

$$T = \alpha \cdot \text{Vol}_k \propto \left[\frac{1-d}{d} \cdot d^2 \right] \propto (1-d) \cdot d$$

$$K_s \propto \frac{(1-d)}{d}$$

$$\text{optimum} = \underline{0.5}$$

$\rightarrow$ electric loading



MAIN FLUX PATHS

Example 3.1 : Carter's Coefficient

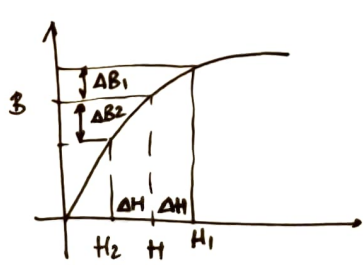
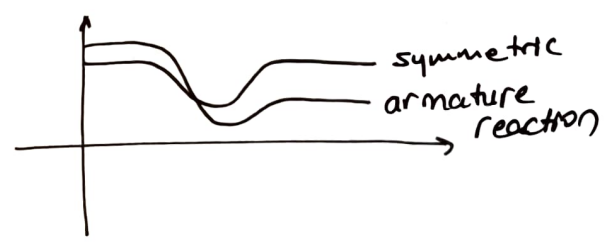
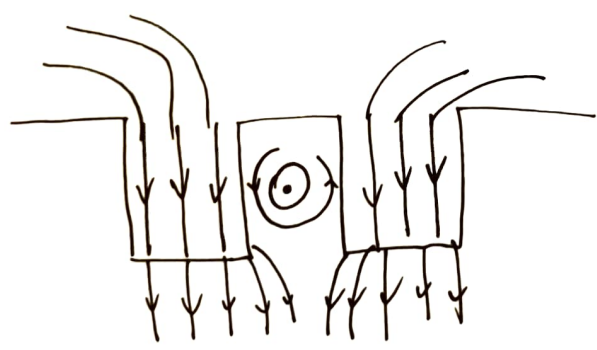
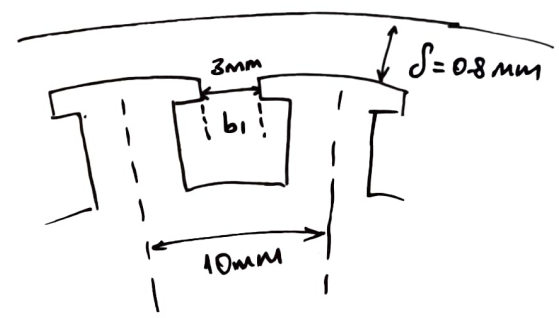
$\delta = 0.8mm$ air gap $b_1 = 3mm$ slot opening $\tau_u = 10mm$ slot pitch

$$K = \frac{b_1 / \delta}{5 + b_1 / \delta} = \frac{\frac{3}{0.8}}{5 + \frac{3}{0.8}} = 0.429$$

$$b_c = K b_1 = 0.429 \times 3 = 1.29mm$$

$$k_c : \text{carter coefficient} \quad k_s = \frac{\tau_u}{\tau_u - b_c} = \frac{10}{10 - 1.29} \approx 1.148$$

$$\delta_{eff} = k_s \times \delta = 0.8 \times 1.148 = \underline{0.918mm} \quad \text{excess part is exercise!}$$

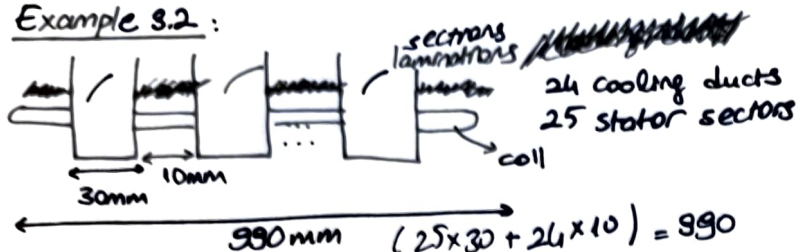


For non linear materials with operating point close to saturation, armature reaction effect dominates further.

Effective Core Length

(16)

Example 3.2 :



$$l_{arr} = 3mm$$

$$K = \frac{\frac{10}{3}}{\frac{5+10}{3}} \approx 0.4 \quad b_{ve} = 0.4 \times 10mm = 4mm$$

$\uparrow$
 b_v

$$l' = 990 - 24 \times \frac{4}{3} + 2.3 \Rightarrow$$

$$l' = 900mm \Rightarrow \text{Actual material is 750mm}$$

Magnetizing Inductance

$$L = \frac{\lambda}{I} = \frac{N\Phi}{I} \Rightarrow \Phi = \int B ds$$

if B is sinusoidal

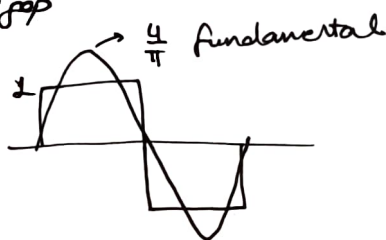
$$\hat{\Phi}_{pole} = \frac{2}{\pi} \hat{B} \tau_{pole} l'$$

$$\lambda = N\Phi = k_w, N_s \hat{\Phi}_{pole}$$

$$\lambda = k_w, N_s \frac{2}{\pi} \hat{B} \tau_{pole} \hat{\lambda}$$

$$\hat{H}_m \delta_e = MMF = \frac{\hat{B}}{\mu_0} \delta_e \rightarrow \text{effective arr gap}$$

$$MMF = \frac{4}{\pi} \frac{k_w, N_s}{2p} \sqrt{2} I_{s,rms}$$



$$\hat{B} = \frac{4 k_w, N_s}{\pi} \frac{\sqrt{2} I_{s,rms}}{2p} \frac{\mu_0}{\delta_e}$$

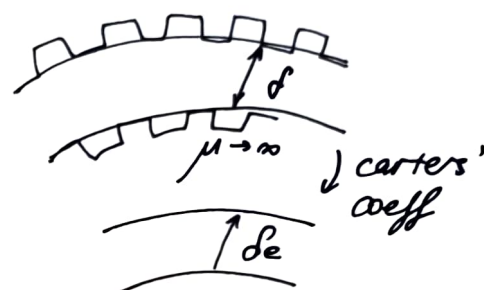
$$\lambda = \frac{2}{\pi} \frac{\mu_0}{\delta_e} \frac{4}{\pi} \frac{(k_w, N_s)^2}{2p} \tau_p l' \sqrt{2} I_{rms}$$

$$L = \frac{\lambda}{I} = \frac{2}{\pi} \mu_0 \frac{1}{2p} \frac{4}{\pi} \tau_p l' (k_w, N_s)^2$$

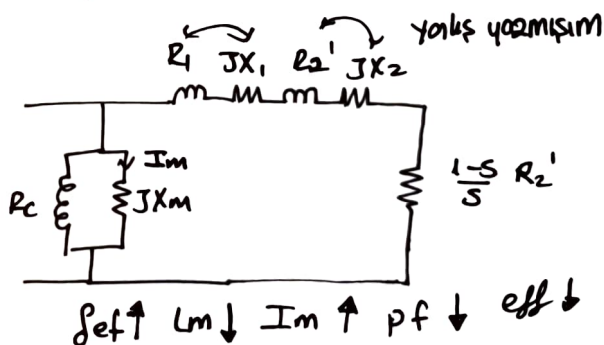
$\uparrow$
pole pitch

$$L_m(ph) = \frac{2\mu_0 D}{p^2 \pi \delta_e} l' (k_w, N_s)^2$$

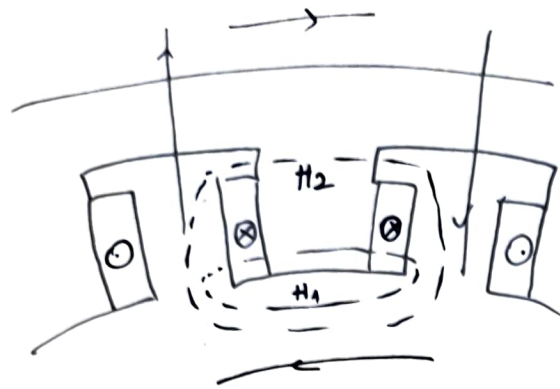
$$L_m = \frac{3}{2} L_m(ph)$$



Induction motor equivalent circuit



Pole Leakage Flux



$$\int H dl = NI$$

$$H_2 \gg H_1$$

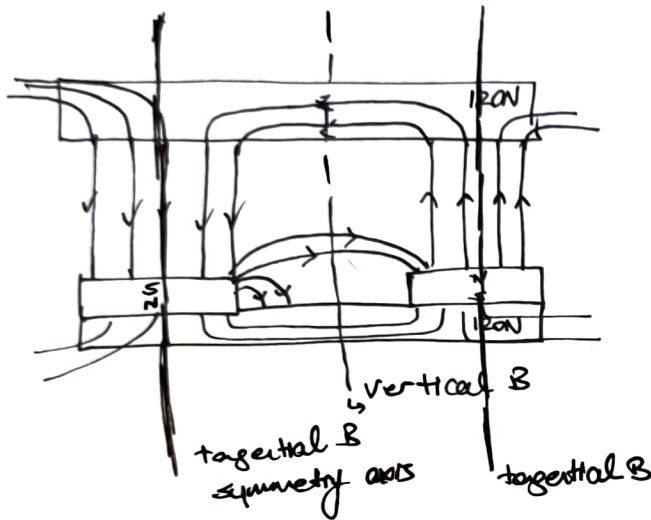
$$\text{as } I_2 \gg I_1$$

$$l_2 \ll l_1$$

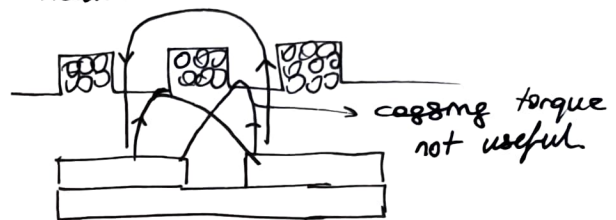
Hemtel

(17)

roket san
yapiskan bilm
tell



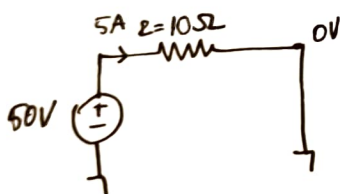
model leakage with reluctance network.



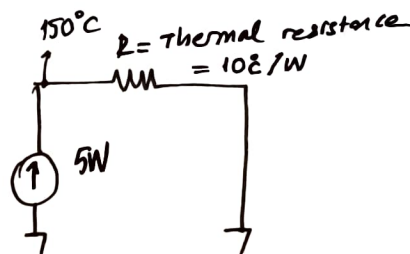
THERMAL MODELING

Conduction

Electric circuit



Lumped Thermal Model



Temp $\equiv$ Voltage

Heat input $\equiv$ Current source

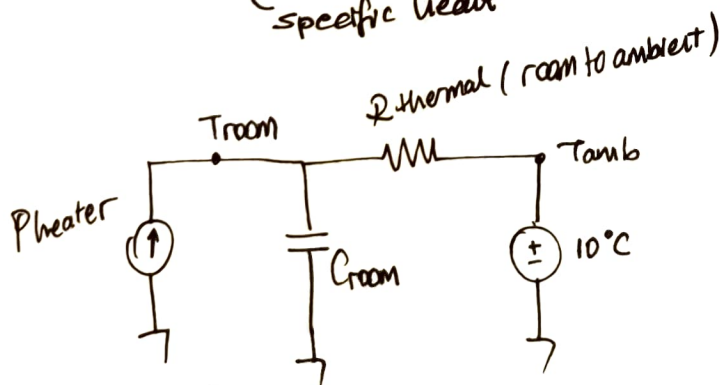
thermal conductivity $\equiv$ Electrical Conductivity

Heat capacity $\equiv$ Capacitance

ΔW : Heat in Joules

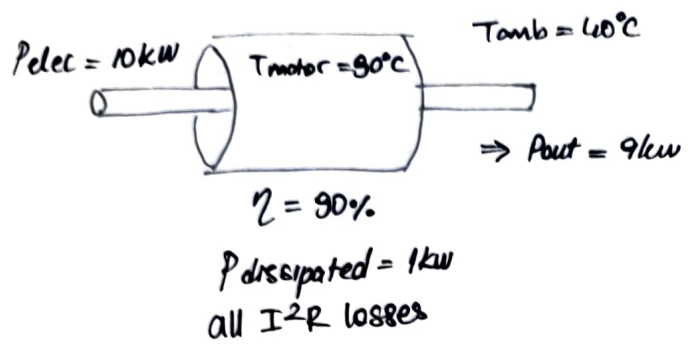
$$\Delta W = m c \Delta T$$

specific heat



$$P = \frac{\Delta T}{R} \quad (\equiv I = \frac{V}{R})$$

Example :

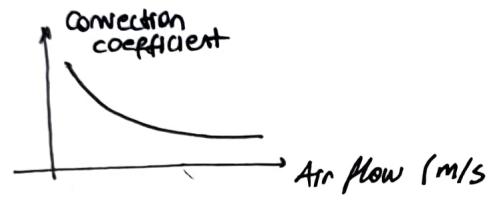


$\rightarrow P_{elec} \% 50 \text{ m\ddot{o}te}$
 $P_{elec} = 15 \text{ kW}$
 $p_{loss} \cdot (1,5)^2 = 2,25 \text{ kW}$
 $\Delta T = 50^\circ \text{C} \times 2,25 = 112,5^\circ \text{C}$
 $T_{motor} = 40 + 112,5 = \underline{\underline{152,5^\circ \text{C}}}$
! Assumption of constant R

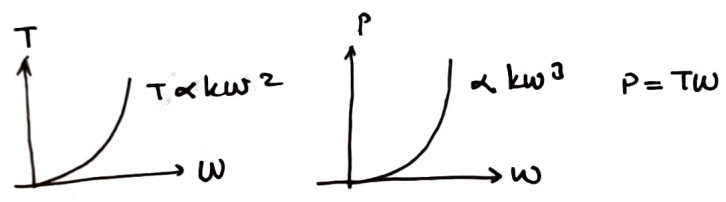
$P = \frac{\Delta T}{R}$

Convection

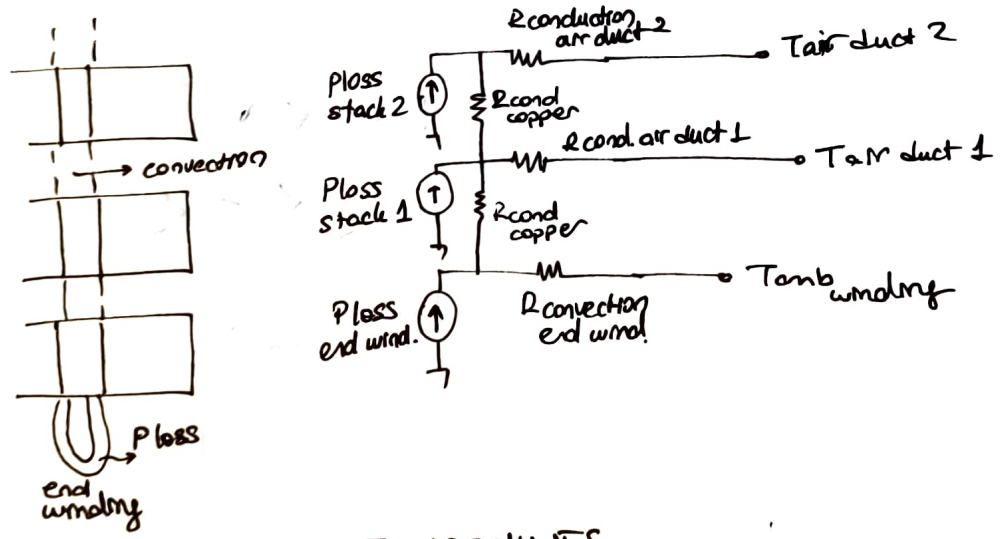
$P = \frac{1}{hA}$



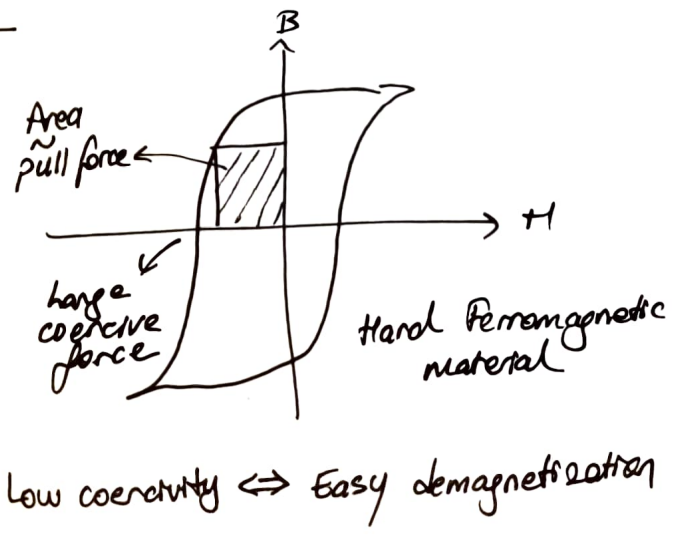
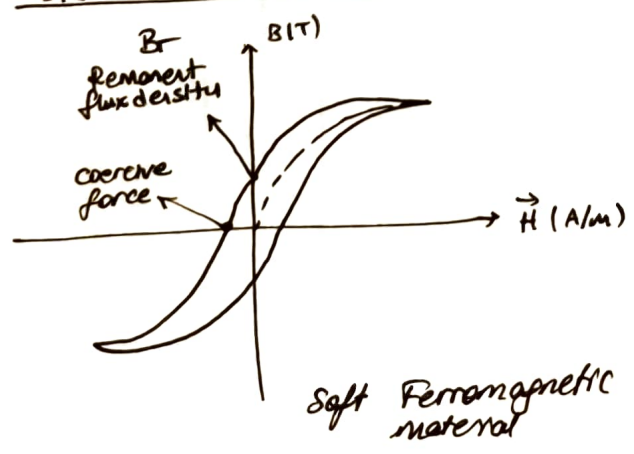
For fan load,

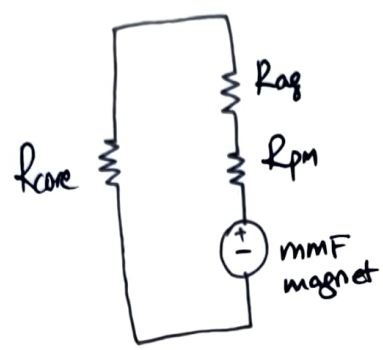
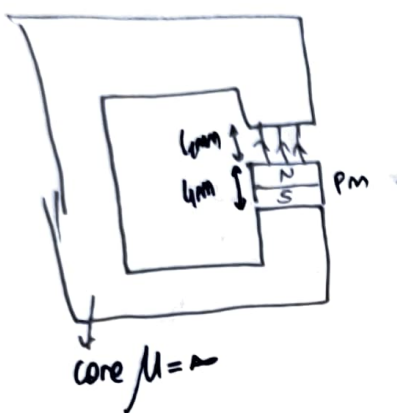


To make the blowing speed $\times 2$, one should make $P \times 3$.



PERMANENT MAGNET MACHINES





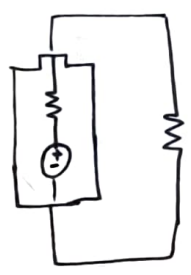
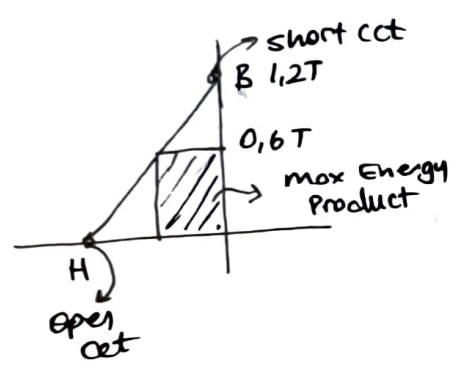
$$R_{air} = \frac{l}{\mu_0 A}$$

$\downarrow$
 $4\pi \times 10^{-7}$

$$R_{pm} = \frac{l}{\mu_0 \mu_r A}$$

$\downarrow$
1.05

4mm magnet $\rightarrow B_{ag} = 0.6T$
8mm magnet $\rightarrow B_{ag} \neq 1.2T$) mmF doubled
 R is also increased! $\Rightarrow$ PM is like air.



$$R_{load} = R_{int} \therefore \text{Max Energy Product}$$